

Clocking / Synchronization Plane

Legend

Data plane

Control plane

Clocking / sync

RF front-end

DDR memory

Processor system

AXI-Stream

AXI-MM data

AXI-Lite ctrl

clock

reset

interrupt

The diagram illustrates the Clocking / Synchronization Plane, showing the flow of clock signals and resets from external inputs to various system components.

External Clock Inputs: Three inputs are shown: `sys_clk_dds4`, `lmk_clk2`, and `lmk_clk1`.

clocktreeMTS: This block contains the core clocking logic:

- `IBUFDS_DDR4` and `IBUFDS_SYSREF` are connected to `sync_0..3 (SYSREF CDC -> MTS)`.
- `IBUFDS_PL_CLK` is connected to `clk_wiz_0 (MMCM)` with a `ref` signal.
- `sync_0..3` is connected to `BUFG_PL_CLK`.
- `clk_wiz_0 (MMCM)` is connected to `BUFG_DDR4`.

Outputs and Connections:

- `BUFG_PL_CLK` outputs `user_sysref` and `rf_clk 245.76`.
- `BUFG_DDR4` outputs `ddr_clk 200`.
- `clk_wiz_0 (MMCM)` outputs a `locked` signal to `binary_latch_counter_0 (locked seq.)`.
- `zynq_ultra_ps_e_0 (pl_clk0 100MHz, pl_resetn0)` outputs `pl_resetn0` and `pl_clk0` to `clk_wiz_0`.

reset_block: This block contains reset logic:

- `binary_latch_counter_0 (locked seq.)` outputs `c0_ddr4_ui_clk` to `ddr4_0`.
- `rst_adc_245M proc_sys_reset` outputs `rf aresetn`.
- `rst_ddr4_200M proc_sys_reset` outputs `ddr4 aresetn`.
- `rst_ps8_100M proc_sys_reset` outputs `ps8 aresetn`.
- `clk_wiz_0` outputs `pl_clk0` to `adc_path / dac_path / ps8_axi_periph`.

Other Components:

- `ddr4_0 (c0_sys_clk_i)` receives `ddr_clk 200` and `c0_ddr4_ui_clk`.
- `usp_rf_data_converter_0 (axis aclk + user_sysref)` receives `user_sysref` and `rf_clk 245.76`.
- `adc_path / dac_path / ps8_axi_periph (aclk + aresetn fan-out)` receives `pl_clk0` and `ddr4 aresetn`.